

GCP Aviation Lakehouse Platform

GCP Resource Inventory | Project: **gcp-lakehouseproject** | Region: **us-central1** | Generated: May 13, 2026

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State Bucket

 Cloud Storage

Defined in storage.tf

Resource Type	Name / ID	Purpose
GCS Bucket	gcp-lakehouseproject-bronze	Raw landing zone — ingest writes CSV here (Stage: SOURCE → BRONZE)
GCS Bucket	gcp-lakehouseproject-silver	Cleaned Parquet data after Databricks transformation (BRONZE → SILVER)
GCS Bucket	gcp-lakehouseproject-gold	Aggregated analytics-ready Parquet (SILVER → GOLD)
GCS Bucket	gcp-lakehouseproject-ai	Vertex AI embedding artifacts and RAG manifests
GCS Bucket	gcp-lakehouseproject-tfstate	Terraform remote state (manually provisioned)

 BigQuery

Defined in bigquery.tf

Resource Type	Name / ID	Purpose
Dataset	aviation_analytics	Top-level dataset containing all tables and views

Resource Type	Name / ID	Purpose
External Table	silver_flights_ext	Parquet-backed external table over silver GCS bucket
External Table	gold_summary_ext	Parquet-backed external table over gold GCS bucket
View	bi_airline_performance_v	BI: per-airline delay and on-time performance
View	bi_route_performance_v	BI: per-route delay statistics
View	bi_daily_delays_v	BI: daily delay trend aggregations
View	bi_pipeline_refresh_v	BI: pipeline freshness and data latency
View	ai_delay_explanations_v	AI: structured delay explanations for RAG context
View	ai_route_risk_v	AI: route risk scoring for RAG retrieval
Native Table	ai_rag_documents	Primary source for RAG — flight records as JSON documents
View	ai_nl_analytics_facts_v	AI: structured facts for natural-language analytics queries

 GKE Autopilot & Artifact Registry

Defined in *gke.tf*

Resource Type	Name / ID	Purpose
GKE Autopilot Cluster	aviation-pipeline	Runs daily ingest CronJob (06:00 UTC) — synthetic flight data generator
Artifact Registry	aviation-pipeline (Docker)	Stores ingest and retrieval service container images
Service Account	aviation-pipeline-sa	GKE Workload Identity SA — scoped to Bronze bucket write only

 Vertex AI

Defined in *vertex_ai.tf*, *vector_search.tf*

Resource Type	Name / ID	Purpose
Service Account	aviation-ai-sa	AI pipeline SA for embedding generation and BigQuery analytics
Vector Search Index	aviation-rag	Stores text-embedding-005 vectors for aviation RAG documents
Index Endpoint	aviation-rag	Public endpoint for semantic nearest-neighbour queries
Models (API)	text-embedding-005 / gemini-2.5-flash	Embedding model and reasoning model — consumed via API, no provisioning

 Cloud Run

Defined in `retrieval_service.tf`

Resource Type	Name / ID	Purpose
Cloud Run Service	aviation-retrieval	Flask HTTP service — RAG retrieval + Gemini reasoning + session memory
Service Account	aviation-retrieval-sa	Cloud Run identity — roles: aiplatform.user, bigquery, storage, datastore.user

 Firestore

Defined in `firestore.tf`

Resource Type	Name / ID	Purpose
Firestore Database	rag-sessions	Durable conversation history store for RAG session memory (Native mode)
TTL Field Policy	sessions.expireAt	Auto-deletes session documents after 1 hour of inactivity

 IAM Roles Summary

Key bindings across all `.tf` files

Service Account	Role	Scope
aviation-pipeline-sa	roles/storage.objectCreator	Bronze bucket only

Service Account	Role	Scope
aviation-pipeline-sa	roles/aiplatform.user	Project
aviation-pipeline-sa	roles/bigquery.user + dataEditor	Project
aviation-ai-sa	roles/aiplatform.user	Project
aviation-ai-sa	roles/bigquery.user + dataViewer	Project
aviation-ai-sa	roles/storage.objectAdmin	AI artifacts bucket
aviation-retrieval-sa	roles/aiplatform.user	Project
aviation-retrieval-sa	roles/bigquery.dataViewer + jobUser	Project
aviation-retrieval-sa	roles/storage.objectViewer	AI artifacts bucket
aviation-retrieval-sa	roles/datastore.user	Project (Firestore)